

Jaineel Modi

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Education

Queen's University

Expected May 2028

Bachelor of Computing - Specialization in AI, GPA: 3.74, Dean's Honour List x3

Kingston, ON

- Courses: Data Structures & Algorithms, Software Architecture, Database Management Systems, Artificial Intelligence
- Activities: Hack the North (University of Waterloo), KingsHacks, AI Society, Competitive Programming Club

Technical Skills

Languages: Python, Java, C#, TypeScript, JavaScript, Ruby, Rust, SQL, Bash

Frameworks/Libraries: Spring Boot, React, Next.js, Node.js, Flask, .NET Core, PostgreSQL, MySQL, MongoDB

DevOps & Tools: AWS, Azure, Docker, Kubernetes, Kafka, Camunda, Git, CI/CD, REST APIs, Jira

AI & Machine Learning: OpenAI API, LangChain, Claude Code, MCP, Hugging Face, PyTorch, TensorFlow

Experience

Software Developer Intern

May 2026 – Present

Royal Bank of Canada (RBC)

Toronto, ON

- Built client-facing **REST APIs** for RBC's mortgage approval platform team, automating debt-servicing, housing cost, amortization, and income-qualification calculations across **10,000** mortgage applications.
- Built a **Spring Boot** microservice that automatically validated mortgage documents against business rules, replacing manual reviews and reducing validation effort by **95%**.
- Built **GitHub Actions** pipelines that automate ingestion and synchronization of Markdown documentation from engineering repositories and Confluence across **8** services, delivering real-time context to internal AI agents.
- Migrated **5** configuration services from **Azure SQL** to **MongoDB** with immutable audit trails for every change.

Student Researcher

Sept 2025 – Jan 2026

Queen's University (RISE Lab)

Kingston, ON

- Developed a method to trace failures in **multi-agent LLM coding systems** under **Prof. Yuan Tian**, identifying which agent and step causes a repair attempt to break.
- Built the repair pipeline in **PyTorch**, logging multi-agent trajectories and training a tracer with counterfactual replay and targeted fault injection, lifting **Qwen2.5-Coder** pass@1 on **HumanEval** by +12.8 pts.

Software Developer Intern

Jun – Aug 2024 & May – Aug 2025

Mulvey & Banani International

Toronto, ON

- Shipped a Revit Copilot powered by **GPT-4** via the **OpenAI API**, automating **20+** BIM tasks with natural-language prompts and multi-step tool execution, serving **80+** users and handling **5,000+** prompts per month.
- Collaborated with the AI team to automate blueprint generation in Revit by placing **250+** electrical components using the **Revit API**, **C#**, and **.NET**, boosting productivity by **20%** and saving **26 hours** per project.
- Deployed **Azure Computer Vision** to automate image tagging, accelerating search and retrieval for **10k+** images.

Software Developer Intern

Jan 2024 – May 2024

ConvoSearch

San Francisco, CA

- Engineered an AI-powered shopping interface by integrating **GPT-4** with few-shot and Chain-of-Thought prompting via **LangChain** to validate and transform client data in an ETL pipeline, achieving **98.3%** ingestion accuracy.
- Implemented a profit-driven reranking layer that boosted CVR by **18%** and increased average order value by **25%**.
- Developed scalable CRUD APIs in **Python** for **MongoDB**, optimizing performance with query indexing and pagination to handle **50k+** records and reduce average execution time by **30%**.

Projects

AudioMark | Flutter, Dart, just_audio, Provider | 🐙

- Co-built a **Flutter** music player for dancers on iOS/Android, live on **Google Play** with **5000+** active users.
- Managed full SDLC from design to **Google Play** release, with automated unit and E2E tests via **GitHub Actions**.

MovieMatch App | SwiftUI, Firebase, Flask, Node.js (Express), PostgreSQL, Recsend, Ruby | 🐙

- Created an **iOS** movie recommendation app with a swipe-based engine that learns user taste from like/dislike signals.
- Fine-tuned **all-MiniLM-L6-v2** on **SNLI/MNLI** with NLI softmax loss, achieving **91.2%** accuracy on validation set.

TossIt 2.0 | Next.js, TypeScript, Tailwind, FastAPI, ONNX, Firebase, VGG16, Claude Code | 🐙

- Awarded **first place** at SatecHack for an ML app that identifies user waste and promotes better recycling habits.
- Trained a **VGG16** image classifier achieving **97%** accuracy in waste categorization, served via **ONNX** and **FastAPI**.